
Software Requirements Specification

for

Autonomous Delivery Cart

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version
Ahmed Sameh	28/9/2025	Initial Draft of SRS	1.0

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for the **Autonomous Delivery Cart System**, a graduation project designed as a proof-of-concept for a future start-up. The system aims to automate last-mile delivery within closed environments such as residential compounds, hotels, or universities. This document specifies the software and cloud requirements, including customer mobile applications, vendor dashboards, and backend cloud infrastructure.

1.2 Document Conventions

This SRS follows IEEE software specification guidelines. Requirements are written in clear, unambiguous statements and identified using the label **REQ-x** for traceability. System features are grouped by user type and functionality. All acronyms are defined in the glossary.

1.3 Intended Audience and Reading Suggestions

This document is intended for the development team (Ahmed: Cloud and Software Architecture, Youssef: Mobile Development), supervisors (Prof. Daa and assistants), and potential evaluators. Developers should focus on the detailed functional and nonfunctional requirements in Sections 3–5. Supervisors should review Sections 1 and 2 for project scope and objectives. Future testers and maintainers should refer to the system features in Section 4 and nonfunctional requirements in Section 5.

1.4 Product Scope

The **Autonomous Delivery Cart System** provides a contactless, efficient, and automated delivery solution within contained environments such as compounds and hotels. Customers can place and track orders via a mobile application, vendors can manage inventory and deliveries through a dashboard, and autonomous carts execute the deliveries. The system aims to reduce manual delivery efforts, improve service speed, and demonstrate a scalable solution for future commercial use.

1.5 References

AWS official documentation: <https://docs.aws.amazon.com>

IEEE SRS standard template (IEEE 830-1998)

Project discussions and requirement gathering sessions (internal team notes, Sept–Oct 2025)

2. Overall Description

2.1 Product Perspective

This system is a new standalone product integrating hardware, cloud, and mobile applications. It does not replace an existing system but introduces automation into the delivery workflow. The product is modular: customers interact via mobile, vendors use a dashboard, and carts are controlled through AWS IoT and backend services.

2.2 Product Functions

Key functions of the system include:

- Allowing customers to register, log in, browse items, place orders, and track deliveries.
- Providing vendors with a dashboard to confirm orders, manage inventory, and dispatch carts.
- Supporting managers with analytics and reporting tools.
- Enabling real-time cart tracking and secure cart-door unlocking via OTP verification.
- Storing sensor and operational data in the cloud for monitoring and analytics.

2.3 User Classes and Characteristics

- **Customers:** Non-technical, use the mobile app occasionally to order goods.
- **Vendors:** Semi-technical, use the web dashboard regularly to manage orders and carts.
- **Managers:** Technical/administrative, need system-level visibility, analytics, and controls.
- **Carts:** Autonomous units acting as system users, requiring secure machine-to-cloud authentication.

2.4 Operating Environment

- **Cloud:** AWS (Cognito, DynamoDB, IoT Core, API Gateway, Lambda, S3).
- **Mobile Application:** Android and iOS devices.
- **Web Dashboard:** Modern browsers (Chrome, Edge, Safari).
- **Cart Hardware:** NVIDIA Jetson TX2/Nano, LiDAR, GPS, Kinect/ZED camera, battery, actuators.

2.5 Design and Implementation Constraints

- Authentication must use AWS Cognito.
- All communication must be secure (TLS).
- Real-time updates are required with latency below 500 ms for cart control.
- Raw sensor data storage should use cost-optimized solutions like AWS S3 Glacier.
- Hardware limitations of Jetson devices require lightweight models and efficient computation.

2.6 User Documentation

- Mobile app user guide.
- Vendor dashboard manual.
- System admin/maintenance guide.
- Technical architecture and deployment documentation.

2.7 Assumptions and Dependencies

- Stable Wi-Fi or 4G/5G connectivity is available in compounds/hotels.
- AWS services are accessible and funded.
- Sensor hardware operates within expected accuracy.
- Payment gateway integration is postponed but expected in future versions.

3. External Interface Requirements

3.1 User Interfaces

- **Customer App:** Login/register, item browsing, cart tracking, delivery OTP entry, notifications.
- **Vendor Dashboard:** Order list, inventory management, cart dispatch, reports.
- **Manager Dashboard:** System analytics, cart status overview, emergency controls.

3.2 Hardware Interfaces

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3.3 Software Interfaces

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3.4 Communications Interfaces

- **Mobile/Web:** HTTPS (REST/GraphQL).
- **Carts:** MQTT (IoT Core).
- **Notifications:** SNS/FCM for push notifications.

4. System Features

4.1 Customer Ordering

Customers can browse items, place orders, and track cart location.

Priority: High

- REQ-1: App must allow item selection and checkout.
- REQ-2: App must provide order status and tracking.

4.2 Vendor Management

Vendors confirm orders and dispatch carts.

Priority: High

- REQ-3: Vendors must approve orders before dispatch.
- REQ-4: Vendors must view completed and pending orders.

4.3 Cart Control

The backend controls carts and secures delivery.

Priority: High

- REQ-5: Carts must receive navigation and status commands securely.
- REQ-6: Cart door must only unlock after OTP verification by the customer.

4.4 Vendor Management

Managers oversee analytics and system health.

Priority: Medium

- REQ-7: System must generate daily/weekly analytics.
- REQ-8: Manager must be able to trigger an emergency stop.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Command latency below 500 ms.
- Mobile response under 2 seconds.
- Support up to 1,000 concurrent users in pilot phase.

5.2 Safety Requirements

- Carts must auto-stop if sensors fail.
- Emergency always stop accessible to managers.

5.3 Security Requirements

- All data must be encrypted in transit and at rest.
- OTP must expire after 5 minutes.
- Each cart must authenticate with unique IoT credentials.

5.4 Software Quality Attributes

- Reliability: 99.9% uptime target.
- Scalability: Horizontal scaling for multiple compounds.
- Usability: Simple design for non-technical users.
- Maintainability: Clear modular design with AWS IaC.

5.5 Business rules

- Customers cannot place orders without authentication.
- Vendors must approve before carts are dispatched.
- Orders are only confirmed upon OTP verification.

Appendix A: Glossary

- **AWS:** Amazon Web Services.
- **OTP:** One-Time Password.
- **IoT Core:** AWS managed MQTT service.
- **Vendor:** Store/pharmacy supplying goods.

Appendix B: Analysis Models

- ERD diagram (provided separately).
- Sequence diagrams (to be developed).

Appendix C: To Be Determined List

- Final payment provider.
- Final list of sensors on cart.
- SLA for compound/hotel network coverage.